

A whole lot of reductions

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1 preliminaries

Definition 1. A **tour** in a graph $G = (V, E)$ is a sequence of vertices and edges adjacent to them in the graph that spans every vertex. We usually refer to the multiset of edges defined by a tour as T . The **cost** of a tour T is the sum of the weights of the edges in T , denoted as $c(T) = \sum_{e \in T} c(e)$.

This definition is equivalent to saying that a tour is a Eulerian multigraph that spans every vertex of G .

Definition 2 (Metric TSP). Let $G = (V, E)$ be a connected undirected graph. The **Metric Traveling Salesperson Problem** (Metric TSP) is defined as follows:

- **Instance:** A complete graph $G = (V, E)$ which is connected by definition, and a cost function $c : E \rightarrow \mathbb{R}_{\geq 0}$ that satisfies the triangle inequality:

$$c(u, v) \leq c(u, w) + c(w, v) \quad \text{for all } u, v, w \in V. \quad (1)$$

- **Objective:** Find a closed tour C in G such that the total weight $c(C) = \sum_{e \in C} c(e)$ is minimized.

Definition 3 (Quasi-Tour). A **quasi-tour** T_0 in a graph $G = (V, E)$ is a subset of edges in T removing edges from a finite set of closed tours. Each connected component of this multigraph is either a Eulerian multigraph or an isolated vertex.

A given multiset of edges T_0 is a quasi-tour if and only if it is Eulerian.

2 Starting point

For the reduction, we will start from a common hard to approximate problem:

Definition 4 (MAX-E3-LIN-2). The **MAX-E3-LIN-2** problem is defined as follows:

- **Instance:** A set of n boolean variables $x_1, x_2, \dots, x_n$ and a collection of m linear equations over $\mathbb{F}_2$, where each equation involves exactly three variables. Each equation is of the form:

$$x_{i_1} + x_{i_2} + x_{i_3} = b_j, \quad (2)$$

where $b_j \in \{0, 1\}$ and i_1, i_2, i_3 are distinct indices from $\{1, 2, \dots, n\}$.

- **Objective:** Find an assignment to the variables that maximizes the number of satisfied equations.

Theorem 1. For every $\varepsilon \in (0, \frac{1}{4})$ and a sufficiently large integer $k \geq k(\varepsilon)$, the following partial decision subproblem $Q(\varepsilon, k)$ of MAX-E3-LIN-2 is NP-hard: given an instance of MAX-E3-LIN-2 with m equations and exactly k occurrences of each variable, to decide if at least $(1 + \varepsilon)m$ equations can be satisfied or if at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied by an optimal assignment.

To prove some inapproximability results, it is more convenient for reductions if all equations have the same right hand side. The results obtained in such case can be enforced by flipping some occurrences of variables ($\tilde{x} = 1 - x$). With that idea in mind, we can define a subproblem of $Q(\varepsilon, k)$.

Definition 5. For $b \in \{0, 1\}$, we define $Q_b(\varepsilon, k)$ as the following partial decision problem:

- **Instance:** An instance of MAX-E3-LIN-2 with m equations of the form $x_{i_1} + x_{i_2} + x_{i_3} = b$, each variable occurring exactly k times negated and k times unnegated.
- **Objective:** Decide if at least $(1 + \varepsilon)m$ or at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied by an optimal assignment.

To continue with the reduction, we will extend the NP-hardness results to a low occurrence version of w -MAX-E3-LIN-2, a weighted system of linear equations over $\mathbb{Z}_2$ with either 2 or 3 variables. We call this a hybrid system of linear equations. Similarly, the task of the w -MAX-E3-LIN-2 problem is to find an assignment that maximizes the total weight of satisfied equations.

3 Reductions to w -MAX-E3-LIN-2

3.1 From $Q(\varepsilon, k)$ to w -MAX-E3-LIN-2

Let $\varepsilon \in (0, \frac{1}{4})$ and $k > 0$ be an integer such that the problem $Q(\varepsilon, k)$ is NP-hard. Let an instance I of $Q(\varepsilon, k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$. let's assume that there is an amplifier $G = (V, E)$ with edge weights $p : E \leftrightarrow \mathbb{R}^+$, with a set of contacts D with $|D| = k$.

Now, there is a gap preserving reduction from $Q(\varepsilon, k)$ to w -MAX-E3-LIN-2 with the amplifier as a parameter. The instance I of $Q(\varepsilon, k)$ is transformed into a hybrid instance J of w -MAX-E3-LIN-2 as follows:

- For each variable $x \in \nu(I)$, we create a copy of the amplifier G , denoted G_x .
 - Inside G_x , the vertices correspond to variables in J and each edge (u, v) represents the equation $u + v = 0$ with weight $p(u, v)$.
 - Contacts in G_x represent the k occurrences of the variable x in the equations of I . Each contact is associated with one occurrence of x in I .
- Every equation in I of the form $x_{i_1} + x_{i_2} + x_{i_3} = b$ also belongs to J with weight 1.